

SkillPolicyAI: Evidence-Grounded Labour-Market Intelligence for Curriculum Renewal, Skills Policy and Workforce Investment

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Project summary

SkillPolicyAI is a proposed open-source decision-intelligence framework that converts job-market evidence into actionable curriculum and policy intelligence. The system will extract skills, tools, qualifications, experience requirements, salary signals and sector trends from job advertisements and labour-market datasets, map them to recognised occupation and skill taxonomies, compare them with university or vocational syllabi, and generate evidence-based recommendations for curriculum renewal, teaching priorities and government workforce investment.

Universities, vocational providers and government agencies frequently make decisions about new subjects, course renewal, workforce funding and skills policy using evidence that is fragmented, delayed or manually interpreted. At the same time, job advertisements contain near-real-time signals about what companies actually demand: programming languages, cloud platforms, domain knowledge, soft skills, certifications, experience levels, tools, work arrangements and sector-specific capabilities. These signals can help answer practical policy questions: Which skills are rising? Which occupations are in shortage? Which course outcomes are outdated? Which regions need training investment? Which subjects should universities add, remove or redesign?

I propose to develop **SkillPolicyAI**, a transparent, auditable system for extracting labour-market skill demand from job data and converting it into policy and curriculum recommendations. The project will integrate job-posting analytics, skill extraction, occupation classification, curriculum mapping, trend modelling, regional shortage analysis and policy utility scoring. The output will be structured evidence reports, dashboards and curriculum-policy recommendations rather than a black-box recommendation engine.

Scientific and policy problem

The central problem is that labour-market demand changes faster than formal curricula and policy cycles. Employers may start demanding skills such as cloud engineering, cybersecurity, generative AI, data governance, bioinformatics, digital forensics, automation, renewable-energy analytics or health informatics before universities and training providers update their subjects. Government investment decisions can also lag behind industry demand because official statistics, surveys and reports are often published periodically, while job-market signals appear continuously.

The proposed project addresses four linked gaps:

1. **Evidence gap:** job advertisements contain rich skill signals, but these signals are noisy, duplicated, biased toward advertised roles and difficult to aggregate reliably.
2. **Curriculum gap:** course learning outcomes and syllabi are often written in academic language that is not directly mapped to employer skill demand.
3. **Policy gap:** governments and institutions need interpretable evidence about which skills, occupations, sectors and regions should receive investment.
4. **Decision gap:** existing labour-market dashboards describe trends, but often do not translate those trends into explicit curriculum actions or policy options.

The core research question is:

Can job-market evidence be transformed into a reproducible, source-aware decision-support framework for curriculum renewal, skills policy and workforce investment?

Overall aim

The overall aim is to develop and validate **SkillPolicyAI**, an evidence-grounded framework that links job advertisements, occupational taxonomies, curriculum documents and policy datasets to support data-informed education and workforce decisions.

Specific goals

1. **Build a labour-market data ingestion pipeline** for collecting and standardising job advertisements, occupation data, shortage lists, salary information and sector reports from authorised public, institutional or licensed sources.
2. **Develop skill and requirement extraction models** to identify technical skills, soft skills, tools, certifications, qualifications, experience requirements, salary bands, location, work mode and industry context.
3. **Map extracted skills to occupation and skill taxonomies** such as ANZSCO/OSCA, the Australian National Skills Taxonomy, O*NET and ESCO.
4. **Develop curriculum-to-market alignment models** that compare employer demand with university and vocational syllabi, learning outcomes and assessment tasks.
5. **Generate policy and curriculum recommendations** by identifying skill gaps, emerging subjects, regional shortages, reskilling priorities and investment options.
6. **Validate the framework** against expert review, historical labour-market trends, known shortage occupations and curriculum-update case studies.

What information will be collected, and from where?

The project is built around multiple evidence layers. The system will not rely on one job board or one source. It will combine job advertisements with official statistics, taxonomies, education documents and policy resources.

Evidence layer	Example data sources	Use in the system
Job advertisements	Authorised job-board data, institutional job feeds, public-sector job portals, university career boards, LinkedIn/SEEK/Indeed/Adzuna-type sources only where terms or APIs allow.	Extract current employer demand: skills, tools, certifications, years of experience, salary, location, work mode, role title and sector.
Occupation and shortage data	Jobs and Skills Australia, Occupation Shortage List, occupation and industry profiles, Australian Bureau of Statistics labour-force data.	Identify occupations in shortage, regional variation, employment levels and policy-relevant workforce pressure.
Skill taxonomies	Jobs and Skills Australia's National Skills Taxonomy, O*NET, ESCO, ANZSCO/OSCA mappings.	Convert noisy job-ad text into standard skill and occupation categories.
Curriculum documents	University course profiles, subject outlines, accreditation documents, learning outcomes, assessment descriptions and vocational training packages where accessible.	Compare what institutions teach with what employers request.
Education and training data	NCVER, university handbooks, micro-credential catalogues, government training packages and accreditation frameworks.	Identify existing training supply and gaps in formal education provision.
Industry and policy reports	Government reports, industry association reports, employer surveys and workforce strategy documents.	Add qualitative and strategic context to job-ad trends.
Salary and work-arrangement signals	Job advertisements, official wage statistics and public labour-market dashboards.	Estimate market value, remote/hybrid trends and workforce attractiveness.

Important data-governance principle

The project will use **authorised, public, institutional or licensed data sources**. It will not depend on violating job-board terms of service, bypassing access controls or collecting unnecessary personal data. The unit of analysis will be job roles, skills, occupations, curricula and policy trends, not individual job seekers.

Proposed system workflow

Step 1: Job-market evidence collection

The system will ingest job advertisements and labour-market records using APIs, downloadable datasets, partner-provided data or permitted public sources. Each job record will be standardised into a structured schema:

job title, employer sector, location, salary, seniority, employment type, work mode, description, required skills, preferred skills, qualifications, certifications, experience, posting date, source

Duplicate advertisements will be removed or clustered, and job titles will be normalised to occupations using classification models and taxonomy lookup.

Step 2: Skill, tool and requirement extraction

The system will extract structured information from job text. The extraction layer will identify:

- **technical skills:** Python, SQL, cloud computing, machine learning, cybersecurity, molecular modelling, data engineering;

- **tools and platforms:** AWS, Azure, Docker, Kubernetes, Git, Tableau, Power BI, R, Linux;
- **soft and professional skills:** communication, project management, teamwork, stakeholder engagement, leadership;
- **certifications:** AWS certification, CISSP, Scrum, Microsoft, Cisco, project-management certificates;
- **experience levels:** graduate, junior, mid-level, senior, principal, manager;
- **education requirements:** degree area, diploma, vocational certificate, PhD, professional accreditation;
- **work context:** hybrid/remote/on-site, full-time/part-time/contract, salary band and sector.

The extraction will combine dictionary matching, taxonomy mapping, named-entity recognition, embedding-based similarity and large-language-model assisted extraction with audit logs and confidence scores.

Step 3: Occupation and skill taxonomy mapping

Extracted skills and job titles will be mapped to standard occupational and skills frameworks. This is essential because raw job postings use inconsistent wording. For example, one job may say “GenAI”, another may say “LLM”, and another may say “prompt engineering”. The taxonomy layer will group these signals under standard categories.

Framework	Role in SkillPolicyAI
ANZSCO / OSCA	Map Australian job titles to occupation classes and support national workforce reporting.
National Skills Taxonomy	Provide an Australian skills language for comparing employer demand, education provision and transferable skills.
O*NET	Provide detailed occupation profiles, task descriptions, knowledge, skills, abilities and work activities.
ESCO	Provide multilingual occupation and skill concepts useful for international comparison.
Institution-specific curriculum taxonomy	Map local course learning outcomes and assessments to extracted labour-market skill categories.

Step 4: Curriculum and syllabus extraction

The system will process curriculum documents from universities, vocational providers and training packages. It will extract:

- course and subject titles;
- learning outcomes;
- weekly topics;
- assessment tasks;
- practical tools taught;
- graduate attributes;
- accreditation or competency requirements.

The system will then compare curriculum content with labour-market demand. For example, if job ads for software engineering increasingly require Docker, Kubernetes, cloud deployment and MLOps, but a curriculum only teaches basic programming, the system will flag a curriculum-to-market gap.

Step 5: Gap analysis and decision intelligence

SkillPolicyAI will compute several interpretable indicators:

Indicator	Meaning
Skill demand score	How frequently and strongly a skill appears across job ads, weighted by recency, salary, seniority and sector.
Skill growth score	Whether demand for a skill is increasing over time.
Curriculum coverage score	Whether a skill is explicitly taught in course learning outcomes, topics or assessments.
Gap score	High employer demand but low curriculum coverage.
Regional shortage score	Whether demand is concentrated in specific states, cities or regional areas.
Policy urgency score	Combines shortage, growth, salary, strategic priority and training gap.
Investment utility score	Estimates which training or curriculum interventions produce the greatest likely benefit per cost.

Step 6: Policy and curriculum recommendation generation

The final layer will translate evidence into actions. Example recommendations may include:

- add a new subject on cloud engineering, AI safety, cybersecurity operations or health-data analytics;
- update an existing programming subject to include Docker, APIs, testing, GitHub Actions and deployment;
- create a short micro-credential for regional employers needing data analytics or digital transformation;
- prioritise government funding for occupations with high shortage and low training supply;
- identify sectors where reskilling has high labour-market utility;
- recommend teaching staff upskilling based on employer demand and future curriculum needs.

The output will include an evidence report, visual dashboard and machine-readable recommendations.

Example use cases

User	Question	System output
University programme director	Which subjects should be updated in an IT or data-science degree?	Skill-gap report showing missing tools, outdated topics and suggested new modules.
Vocational training provider	Which micro-credentials should we offer next year?	Regional demand ranking for short courses and certifications.
Government policy unit	Where should funding be invested to reduce skill shortages?	Occupation and region-level investment priority map.
Department head	Which teaching skills should staff develop?	Staff capability map compared with emerging job-market requirements.
Accreditation body	Does the curriculum match current industry expectations?	Evidence-based mapping between learning outcomes and employer skill demand.
Student career service	Which skills should students add to improve employability?	Role-specific skill roadmap and missing-skill recommendations.

Connection to my existing work

This idea connects my technical background in AI, evidence extraction, benchmarking and teaching with a broader policy and education problem.

Existing work / experience	Role in SkillPolicyAI
AI-assisted literature extraction and ALR-Bench-style workflows	Provides experience in information retrieval, document ranking, evidence extraction and report generation.
Research-agent / JARVIS-style workflow orchestration	Provides the ability to connect APIs, extract structured information, generate audit logs and produce reports.
GeneMapKit and database harmonisation experience	Transfers to skills and occupation harmonisation, where noisy names must be mapped to standard identifiers.
GWASRanker / GWAS Poker-style quality-aware ranking	Provides methodological experience in ranking data sources by quality, compatibility and downstream usefulness.
Teaching across programming, cybersecurity, digital forensics, mobile development and IT subjects	Provides direct domain understanding of curriculum design, skills teaching and employability gaps.
Research software development	Supports the creation of a reproducible open-source decision-support platform rather than a one-off analysis.

Research design and methods

Aim 1: Build the labour-market evidence dataset

A dataset of job advertisements and official labour-market indicators will be constructed. Each record will be normalised by date, region, occupation, sector, salary, seniority and source. The pipeline will include duplicate detection, job-title normalisation, skill extraction and quality flags.

Aim 2: Develop skill extraction and taxonomy mapping models

The project will compare rule-based extraction, dictionary matching, transformer embeddings, supervised classifiers and large-language-model assisted extraction. Extracted skills will be mapped to structured taxonomies such as the National Skills Taxonomy, O*NET and ESCO. The models will be evaluated using manually annotated job-ad samples and taxonomy-alignment accuracy.

Aim 3: Build curriculum-to-market alignment models

Course profiles, learning outcomes and assessment descriptions will be processed using the same skill extraction framework. The system will compute coverage and gap scores for each course, programme or training package.

Aim 4: Develop policy utility and investment-priority scoring

The project will create interpretable policy indicators, including demand growth, shortage severity, curriculum gap, salary signal, regional importance, strategic priority and implementation feasibility. These will be combined into transparent decision-support scores rather than opaque black-box decisions.

Aim 5: Validate recommendations through case studies

The framework will be tested through three case studies:

1. **Digital and AI skills:** software, data science, cybersecurity, cloud and AI roles.
2. **Health and biomedical data skills:** bioinformatics, clinical data, health informatics and digital health roles.

3. **Regional workforce policy:** occupations with shortages across Australian states or regions.

Validation strategy

Validation level	Evaluation approach
Skill extraction accuracy	Manual annotation of job ads and comparison with extracted skills, tools and qualifications.
Occupation mapping accuracy	Compare model-assigned occupations with human-coded ANZSCO/OSCA or O*NET-style labels.
Taxonomy alignment	Measure how well extracted skills map to National Skills Taxonomy, O*NET and ESCO categories.
Curriculum mapping quality	Ask programme directors or teaching staff to review whether mapped curriculum gaps are accurate.
Shortage detection	Compare predicted shortage/growth signals with official shortage lists and labour-market statistics.
Historical back-testing	Train on older job data and test whether the system predicts later skill growth or shortage signals.
Policy usefulness	Expert review by educators, workforce planners or policy analysts using structured evaluation rubrics.
Robustness and bias	Test sensitivity to job-board source, region, occupation, duplicate ads and over-represented industries.

Expected outputs

The project will produce:

1. an open-source labour-market evidence extraction pipeline;
2. a standardised job-skill-occupation dataset for selected case studies;
3. a skill extraction and taxonomy mapping model;
4. a curriculum-to-market alignment framework;
5. policy and curriculum gap indicators;
6. prototype dashboards and evidence reports for educators and policy makers;
7. a manuscript on labour-market skill extraction and curriculum alignment;
8. a manuscript on data-driven skills policy and workforce investment decision support;
9. reusable teaching and curriculum analytics templates.

Feasibility, ethics and risks

Ethical and governance position

SkillPolicyAI will analyse job advertisements, occupational data, curricula and policy documents. It will not profile individual job seekers or employees. Any employer-level reporting will be aggregated or anonymised where appropriate. If institutional curriculum documents or proprietary job-board feeds are used, access permissions and data-governance requirements will be followed.

Risk	Mitigation
Job ads are noisy and duplicated	Deduplicate advertisements, cluster near-duplicates and use source-quality flags.
Job boards restrict scraping	Use APIs, licensed datasets, public-sector data, institutional feeds or permitted sources only.
Skills are described inconsistently	Use taxonomy mapping, aliases, embeddings and human-reviewed skill dictionaries.
Job ads over-represent certain sectors	Combine job ads with official statistics, shortage lists and industry reports.
Curriculum documents are inconsistent	Standardise learning outcomes, topics and assessments into comparable units.
Policy recommendations may overclaim certainty	Report recommendations as evidence-informed options with uncertainty and assumptions.
Bias across regions or industries	Evaluate source bias and report confidence separately by region, sector and occupation.

Fit with possible host groups

This project is suitable for host groups in public policy analytics, education technology, computational social science, labour economics, human-centred AI, data science, digital transformation, workforce planning or decision-support systems.

The project can be adapted to the host group:

- with a public policy group, the emphasis would be skills investment, regional workforce planning and government decision-making;
- with an education technology group, the emphasis would be curriculum alignment, teaching skills and graduate employability;
- with a computational social science group, the emphasis would be labour-market dynamics, skill networks and occupation transitions;
- with an AI/data-science group, the emphasis would be NLP extraction, taxonomy mapping, explainable ranking and decision intelligence.

Indicative timetable

Period	Planned activities
Months 1–3	Finalise host-aligned use cases; define data-governance plan; collect initial job, taxonomy and curriculum datasets.
Months 4–6	Build ingestion pipeline, job schema, duplicate detection and occupation normalisation.
Months 7–9	Develop skill, tool, certification and experience extraction models; create manually annotated evaluation sample.
Months 10–12	Map skills to taxonomies; construct skill-demand and growth indicators; validate extraction and taxonomy alignment.
Months 13–15	Process curriculum and syllabus documents; build curriculum-to-market alignment and gap indicators.
Months 16–18	Develop policy utility scoring, regional shortage analysis and investment-priority reports.
Months 19–21	Complete case studies in digital/AI skills, health-data skills and regional workforce policy; obtain expert feedback.
Months 22–24	Finalise open-source release, dashboards, documentation, manuscripts and future funding proposal.

Why this project matters

SkillPolicyAI converts job-market signals into practical decisions. It can help universities decide which subjects to update, teachers decide which capabilities to teach, students understand employability pathways, and governments decide where training investment has the strongest evidence base. The project is not simply a job-ad scraper; it is a decision-support framework that links employer demand, occupation taxonomies, curriculum design and policy utility.

The long-term vision is to make workforce and curriculum planning more evidence-grounded. Instead of asking only “what jobs are advertised?”, the framework asks:

Which skills are emerging, which curricula are misaligned, which occupations are in shortage, and which education or policy interventions are most useful?

Selected references and data resources

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